SOLUTION OVERVIEW

Credit Risk Portfolio Analytics | Anshul Pundhir (anshulpundhirwork@gmail.com)

1. Risk Analyst Guide (Dashboard Views)

Dashboard View	Core Workflow & Interactive Analytical Capabilities
Portfolio Overview	Monitors executive KPIs ($601.1M total exposure, 16.10% portfolio default rate), animated risk tier composition, interactive dimensional profiler (Loan Type, Region, Income, Employment, Age), and Income x Loan Type default heatmaps.
Account Deep-Dive	Single-loan diagnostic view. Features a predicted PD speedometer gauge, Basel risk tier badge, local **TreeSHAP** attribution waterfall of key risk factors, borrower financial profile, and actual 12-month repayment DPD timeline.
Anomaly Queue	Proactive pre-bureau outlier triage queue powered by dual-lens scoring. Enables risk officers to record triage notes and persist Confirm/Dismiss decisions with instant live KPI updates.
Strategic Insights	Model governance benchmark comparing Logistic Regression vs LightGBM Champion with ROC/PR curves, feature importance ranking, and strategic risk concentration analytics.

2. AI Governance Matrix (Triggers, Failure Modes & HITL)

AI Feature	Trigger	False Positives (FP) & False Negatives (FN)	Human-in-the-Loop (HITL) Workflow
Ask the Portfolio (Grounded LLM)	Analyst queries the chat input.	**FP:** Ambiguous term mapped to wrong metric (Mitigation: Always displays formula badge & active cohort pill). **FN:** Valid query rejected as out-of-scope (Mitigation: Strict intent router with offline fallback).	The analyst must visually inspect the generated provenance metadata card (sample count, formula, source tables) before utilizing figures in credit committee reporting.
Anomaly Detection (Dual-Lens)	Batch run on data refresh.	**FP:** Benign seasonal cash-flow delay flagged as an anomaly (Mitigation: UI shows peer cohort median context). **FN:** Gradual, peer-conforming decay missed (Mitigation: Caught by primary LightGBM PD model).	Interactive triage queue: Risk analyst reviews the account profile, enters investigative notes, and saves absolute Confirm/Dismiss decisions to an audit disk.

3. Model Performance Scorecard

Model	AUC	KS Stat	Gini	Key Modeling Approach & Validation
Logistic Regression (Baseline)	0.6620	24.78%	0.3240	Standard-scaled continuous + one-hot categorical features. L2 regularized benchmark scorecard for baseline regulatory compliance and pure linear explainability.
LightGBM (Champion)	0.6657	25.57%	0.3314	Depth-constrained gradient boosted decision trees with early stopping and L1/L2 penalties. High non-linear interaction capture with robust interpretability via TreeSHAP.

4. Top 3 Portfolio Insights & Strategic Actions

Strategic Finding & Risk Concentration	Actionable 'So What' (Credit Policy Recommendation)
1. Mortgage Concentration Risk Mortgages hold 52.3% of portfolio exposure ($314.5M) with a severe, portfolio-peak 24.65% default rate.	Impose hard 75% LTV origination caps, mandate semi-annual collateral reappraisals, and increase risk-weighted capital reserves for high-balance loans.
2. Monotonic Income Inversion 24.59% default rate in Low-Income (<$48.5k) vs 9.70% in High-Income, representing a massive 2.5x disparity.	Enforce binding debt-to-income caps (max. 35% for lower-income applicants) rather than relying solely on static, historical bureau credit scores.
3. Low-Income Mortgage Squeeze Low-Income Mortgages suffer a staggering 35.41% cross-cut default rate (>2.2x portfolio baseline).	Discontinue automated underwriting approvals for this segment; mandate 6-month debt service escrow liquidity buffers prior to origination disbursement.

5. Key Assumptions & IT Production Requirements

- **Assumptions:** Synthetic loan distributions map linearly to real retail banking books; monthly repayment grain reflects true closing-cycle snapshots; external credit bureau updates incur a standard 30-day reporting lag.
- **Automated Nightly ETL:** Scheduled orchestration from core banking systems (Finacle/Temenos) to a Snowflake/Databricks feature store with upstream schema validation contracts.
- **Enterprise VPC LLM Gateway:** Private cloud endpoint (Google Vertex AI / Azure OpenAI) with managed secrets, zero data retention, and strict customer PII redaction filters.
- **Persistent Audit Store & MLflow:** A PostgreSQL relational database capturing all analyst anomaly triage decisions (for Basel III/IFRS 9 compliance), and an MLflow model registry with automated monthly retraining triggers when KS drops >3% or PSI exceeds 0.20.